



Rossmoyne Senior High School

Semester One Examination, 2023

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section Two: Calculator-assumed

If required by your examination administrator, please place your student identification label in this box

WA student number: In figures

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In words

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Circle your Teacher's Name:

Mrs Alvaro	Mr Bartoshefski	Mr Buckland	Mr Liu
Mr Lockyer	Mr Luzuk	Mr Mohammadi	
Ms Narendranathan	Mr Ng		
Mr Stephens	Mr Tanday		

Time allowed for this section

Reading time before commencing work:
Working time:

ten minutes
one hundred minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators, which can include scientific, graphic and Computer Algebra System (CAS) calculators, are permitted in this ATAR course examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	53	35
Section Two: Calculator-assumed	12	12	100	101	65
Total					100

Instructions to candidates

- The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
- Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
- You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
- Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
- It is recommended that you do not use pencil, except in diagrams.
- Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
- The Formula sheet is not to be handed in with your Question/Answer booklet.

Markers use only		
Question	Maximum	Mark
8	7	
9	7	
10	8	
11	9	
12	7	
13	9	
14	7	
15	10	
16	9	
17	9	
18	8	
19	11	
S2 Total	101	
S2 Wt ($\times 0.6633$)	65%	

Section Two: Calculator-assumed

65% (101 Marks)

This section has **twelve** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 8

(7 marks)

- (a) Given that $f(x) = 4x^2 + 3x + 1$, determine $f\left(-\frac{1}{3}\right)$. (1 mark)

- (b) State the natural domain and range for the following functions.

(i) $f(x) = \frac{3}{x+4}$. (2 marks)

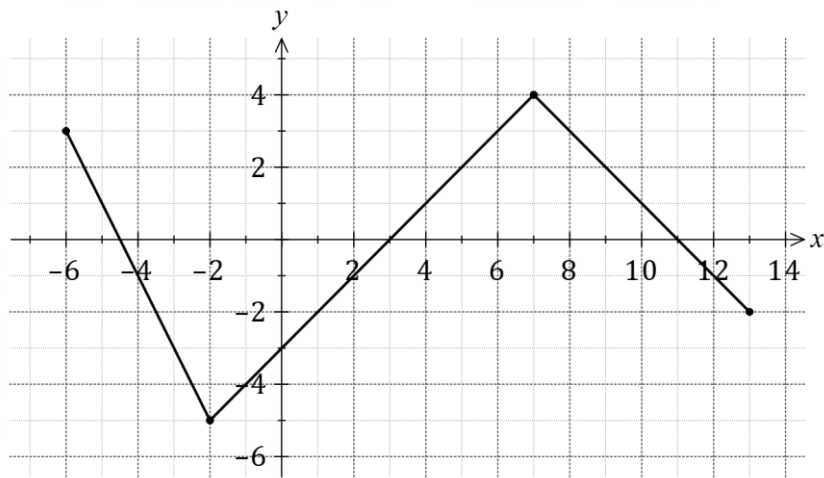
(ii) $g(x) = 5 - \sqrt{x-2}$. (2 marks)

- (c) Determine the range of the function $f(x) = (x-1)^2 - 3$ for the domain $0 \leq x \leq 4$. (2 marks)

Question 9

(7 marks)

The diagram below shows the graph of $y = f(x)$, where f is a function with domain $-6 \leq x \leq 13$.



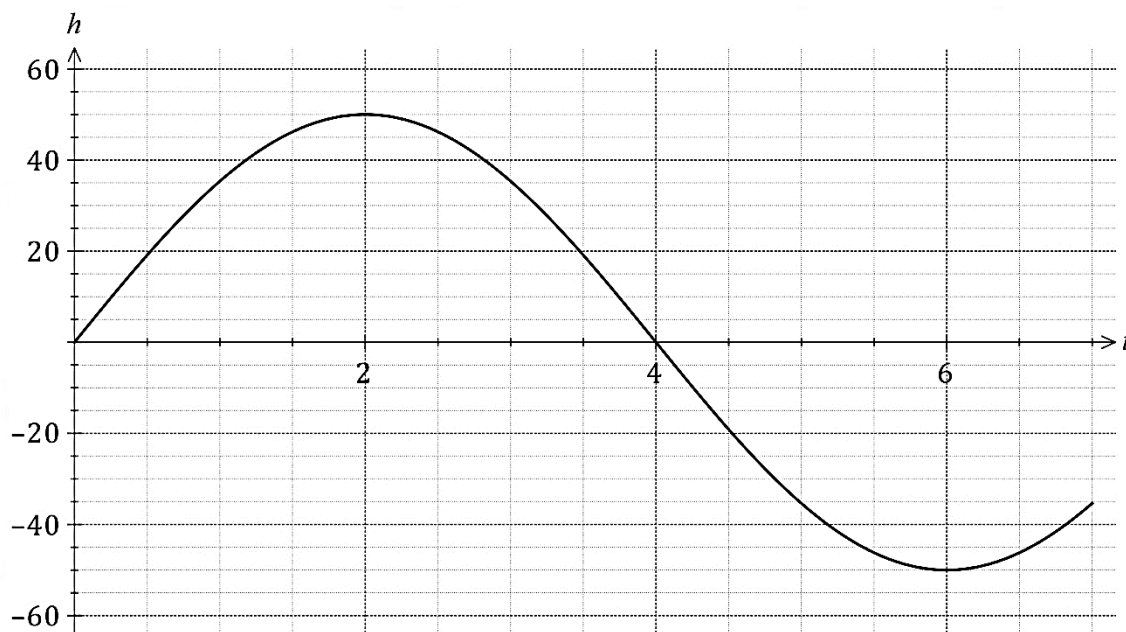
Determine the **coordinates** of the following point(s):

- (a) the y -axis intercept of the graph of $y = f(x - 4)$. (1 mark)
- (b) the roots of the graph of $y = f(x) + 1$. (2 marks)
- (c) the maximum of the graph of $y = -4f(x)$. (2 marks)
- (d) the minimum of the graph of $y = f\left(\frac{x}{2}\right) - 7$. (2 marks)

Question 10

(8 marks)

The vertical height h , in metres, of the tip of a wind turbine blade above its centre of rotation at time t seconds, $t \geq 0$, is shown in the graph below.



- (a) State the amplitude and period of the trigonometric function h . (2 marks)

A similar function for a second wind turbine blade is given by $h_2 = 60 \sin\left(\frac{2\pi t}{3}\right)$, $t \geq 0$.

- (b) State the period of this second function and hence determine how many revolutions per minute this blade is rotating at. (2 marks)
- (c) Add the graph of function h_2 to the axes above. (3 marks)
- (d) At what time does the tip of the second blade first reach its maximum height? (1 mark)

Question 11**(9 marks)**

A dodecahedral dice has 12 sides, each numbered with a different integer between 1 and 12 inclusive. When the dice is rolled once, event A occurs if the number is even, event B occurs if the number is less than 8 and event C occurs if the number is a multiple of 3.

(a) List the sets A , B , and C when the dice is thrown once. (1 mark)

(b) Write a brief word description of the numbers in the outcome set $B \cap \bar{A}$. (1 mark)

(c) List the following sets of outcomes:

(i) $B \cup C$ (2 marks)

(ii) $A \cap \bar{B}$. (1 mark)

(iii) $\overline{A \cup B \cup C}$. (1 mark)

(iv) $A \cap B \cap C$. (1 mark)

(d) Determine $P(\overline{A \cap B} \cap C)$. (2 marks)

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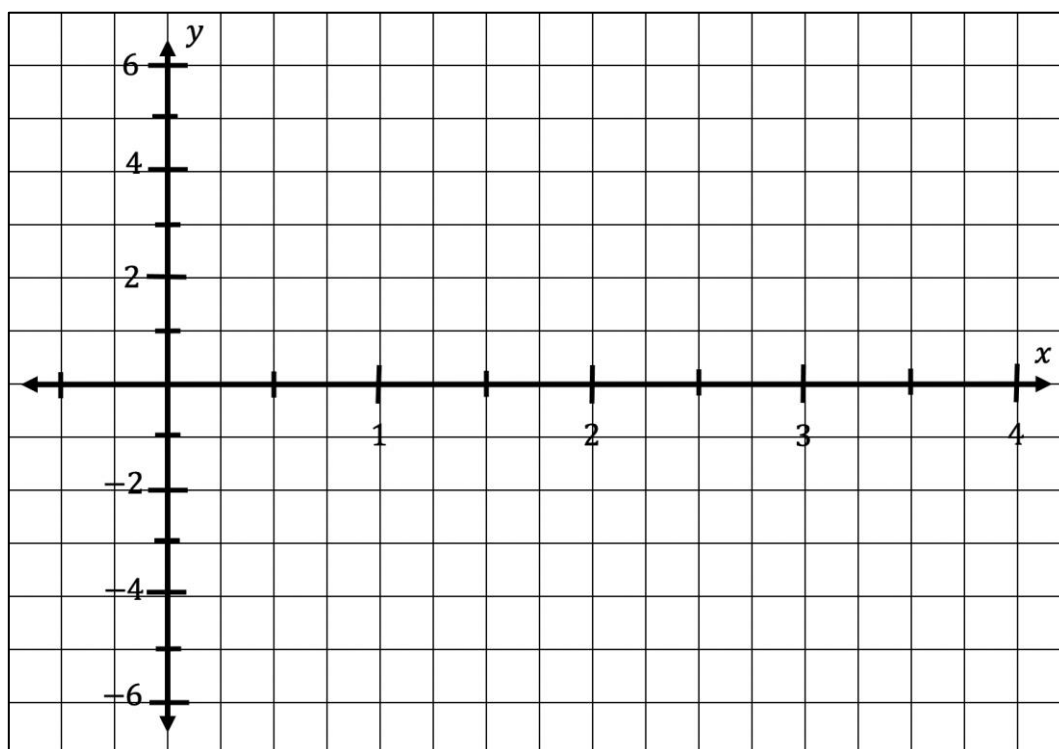
Question 12

(7 marks)

Let $f(x) = x^3 - 6x^2 + 9x - 4$.

- (a) Sketch the graph of $y = f(x)$ on the axes below.

(3 marks)



Let L_1 be the straight line that passes through the local minimum and local maximum of the graph of $y = f(x)$.

- (b) Determine the equation of L_1 .

(2 marks)

Let L_2 be the straight line through $(0, -22)$ and parallel to L_1 .

- (c) Determine x -coordinate of the point of intersection of the graph of $y = f(x)$ and L_2 .

(2 marks)

Question 13

(9 marks)

The graph of the hyperbola $y = \frac{p}{x+q}$, where p and q are constants, passes through the point $(7, 1)$ and intercepts the vertical axis at $y = -\frac{4}{3}$.

- (a) Show how to use this information to deduce that $p = 4$ and $q = -3$. (2 marks)

- (b) State the equations of the asymptotes of the hyperbola. (2 marks)

The line with equation $y = 16x - 48$ intersects the hyperbola at two points.

- (c) Show how to use the equations of the line and the hyperbola to formulate a quadratic equation in the form $ax^2 + bx + c = 0$, the solutions of which are the x -coordinates of the points of intersection. Do **not** determine these x -coordinates. (3 marks)

- (d) Determine the coordinates of the points of intersection of the line and hyperbola.
(2 marks)

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Question 14**(7 marks)**

The table below shows the results of a random survey of 200 customers who recently bought a new TV from a large store. The customers were asked which brand of TV they bought and whether they would buy the same brand again.

Buy same brand again	TV Brand			
	Sense	Sunny	GL	Other
Yes	14	31	19	24
No	11	15	8	30
Unsure	7	6	10	25

A customer who recently bought a TV from the store is selected at random. Use the above data to determine a point estimate for the probability that

- (a) the TV they bought was a Sunny and that they would not buy the same brand again. (1 mark)
- (b) they are unsure whether they would buy the same brand of TV again. (1 mark)
- (c) the brand of TV they bought was not GL. (1 mark)
- (d) they would not buy the same brand of TV again, given that the brand they bought was not 'Other'. (2 marks)

- (e) the brand of TV they bought was Sense or GL, given that they would buy the same brand again. (2 marks)

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Question 15**(10 marks)**

- (a) Determine the area of triangle OPQ when $\angle POQ = 134^\circ$, $OP = 19$ cm and $OQ = 24$ cm.
(2 marks)

- (b) Triangle OXY has an area of 175 cm^2 and has sides $OX = 43$ cm and $OY = 25$ cm. Determine the size of angle $\angle OXY$, giving your answer(s) in degrees correct to one decimal place.
(8 marks)

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Question 16**(9 marks)**

Records kept by a café on an industrial estate show that 74% of its customers work in transport and the remainder in other jobs. 45% of the customers who work in transport order chips with their meal, whilst for other customers the figure is 30%.

- (a) Determine the probability that a customer who arrives at random at the café
- (i) does not work in transport and orders chips with their meal. (2 marks)
 - (ii) does not order chips with their meal. (3 marks)
 - (iii) works in transport given that they order chips with their meal. (2 marks)
- (b) Explain whether the records kept by the café indicate that working in transport and ordering chips with a meal are independent events for their customers. (2 marks)

Question 17**(9 marks)**

A newsagent has a stock of 36 greeting cards, all different to each other, of which 10 are birthday, 11 are anniversary and the remainder are blank cards.

- (a) Determine the number of different selections of three blank cards that could be made from the newsagent. (2 marks)
- (b) Determine the number of different selections of six cards that could be made from the birthday and anniversary cards stocked by the newsagent. (2 marks)
- (c) The newsagent just sold two of the 36 greeting cards.
- (i) Determine the number of different selections of two cards that could be sold. (1 mark)
- (ii) Determine the probability that both were birthday cards. (2 marks)
- (iii) Determine the probability that at least one of them was an anniversary card. (2 marks)

Question 18**(8 marks)**

When a chance experiment is carried out, events X and Y occur with probabilities 0.75 and 0.52 respectively.

(a) Explain why X and Y cannot be mutually exclusive events. (2 marks)

(b) Determine the following probabilities:

(i) $P(\bar{X} \cap Y)$ when $P(X \cap \bar{Y}) = 0.44$. (2 marks)

(ii) $P(X \cup Y)$ when X and Y are independent events. (2 marks)

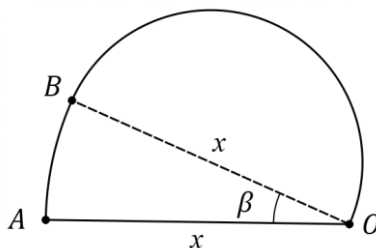
(iii) $P(\overline{X \cup Y})$ when $P(Y | X) = 0.54$. (2 marks)

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Question 19

(11 marks)

The figure shown in the diagram consists of sector OAB of circle with centre O and a semi-circle with diameter OB .



Let $OA = OB = x$ and $\angle AOB = \beta$ radians.

(a) Let $x = 12$ cm and the area of the semi-circle be **three** times the area of the sector.

(i) Show algebraically that β is equal to $\frac{\pi}{12}$. (3 marks)

(ii) Determine the perimeter of the figure. (3 marks)

- (b) By first determining a new value for β , determine the area of the figure when the area of the semi-circle is **five** times the area of the sector and the new perimeter of the figure is 77.2 cm. (5 marks)

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Supplementary page

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